The Road to

Gold

Mathematics for Competitions

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Preface

This book is intended to help students prepare for mathematical competitions, from middle- and high-school olympiads to undergraduate competitions. It brings together mathematical concepts, results, and problem-solving methods that are useful in approaching challenging competition problems.

The book is organized into two parts. *Mathematical Concepts* collects definitions, theorems, formulas, and other mathematical results that can be applied in competitions, from elementary foundations to more advanced theory. These are presented without proofs so that the essential statements and their conditions remain easy to consult. *Problem-Solving Strategies* is devoted to methods of proof, mathematical ideas, and techniques for solving difficult problems, including instructive proofs where appropriate.

This book is a companion to *Mathematics for Science Quiz*, which places greater emphasis on important results, memorable formulas, mathematical history, and interesting facts and connections. The two books may share fundamental definitions and theorems, but they emphasize different purposes: that volume offers broad mathematical knowledge, while this one focuses on the tools and reasoning used in mathematical competitions. Readers may consult either volume independently or use them together.

I hope this book helps readers develop mathematical insight, creativity, and perseverance, and encourages them to aim for the highest achievements in mathematical competitions.

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Part I

Mathematical Concepts

Chapter 1

Algebra

Formula 1.1. Basic Factorization Identities
For elements of a commutative ring,

$$a^2 - b^2 = (a - b)(a + b), \qquad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2),$$

where the two signs in the second identity are paired consistently. More generally, for $n \geq 1$,

$$a^n - b^n = (a - b) \sum_{k=0}^{n-1} a^{n-1-k} b^k.$$

If n is odd, then

$$a^n + b^n = (a + b) \sum_{k=0}^{n-1} (-1)^k a^{n-1-k} b^k.$$

Formula 1.2. Three-Variable Cubic Identity
For a, b, c in a commutative ring,

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca).$$

In particular, $a + b + c = 0$ implies $a^3 + b^3 + c^3 = 3abc$.

Formula 1.3. Square and Fourth-Power Sums
For real a, b, c,

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}\big((a - b)^2 + (b - c)^2 + (c - a)^2\big),$$

$$a^4 + b^4 + c^4 - \sum_{\text{cyc}} a^2 b^2 = \frac{1}{2}\big((a^2 - b^2)^2 + (b^2 - c^2)^2 + (c^2 - a^2)^2\big).$$

Here $\sum_{\text{cyc}} a^2 b^2 = a^2 b^2 + b^2 c^2 + c^2 a^2$.

Definition 1.4. Symmetric and Homogeneous Polynomials
A polynomial $P(x_1, \ldots, x_n)$ is *symmetric* if it is unchanged by every permutation of its variables. It is *homogeneous of degree d* if every nonzero monomial has total degree d, equivalently

$$P(tx_1, \ldots, tx_n) = t^d P(x_1, \ldots, x_n)$$

for every scalar t in an infinite coefficient field.

Formula 1.5. Elementary Symmetric Polynomials
For $x_1, \dots, x_n$ and $0 \le k \le n$, put $e_0 = 1$ and

$$e_k = \sum_{1 \le i_1 < \cdots < i_k \le n} x_{i_1} \cdots x_{i_k}.$$

Then

$$\prod_{i=1}^{n} (t - x_i) = \sum_{k=0}^{n} (-1)^k e_k t^{n-k}.$$

Theorem 1.6. Fundamental Theorem of Symmetric Polynomials
Every symmetric polynomial in $x_1, \dots, x_n$ over a commutative ring R can be expressed uniquely as a polynomial in $e_1, \dots, e_n$ with coefficients in R.

Formula 1.7. Quadratic Equation
For $a, b, c \in \mathbb{C}$ with $a \neq 0$, the solutions of $ax^2 + bx + c = 0$ are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The discriminant is $\Delta = b^2 - 4ac$. For real coefficients, its sign determines whether there are two distinct, one repeated, or no real roots.

Formula 1.8. Two-Variable Symmetric Expressions
If $s = x + y$ and $p = xy$, then x, y are roots of $t^2 - st + p$, and

$$x^2 + y^2 = s^2 - 2p, \quad x^3 + y^3 = s^3 - 3ps, \quad (x - y)^2 = s^2 - 4p.$$

Theorem 1.9. Vieta's Relations for Two and Three Roots
If x, y are roots of $t^2 - st + p$, then $x + y = s$ and $xy = p$. If x, y, z are roots of $t^3 - at^2 + bt - c$, then

$$x + y + z = a, \quad xy + yz + zx = b, \quad xyz = c.$$

Formula 1.10. Sums of Powers of Integers
For $n \in \mathbb{N}$,

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6},$$

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Definition 1.11. Polynomial, Degree, and Multiplicity
For a field F, a polynomial $f(x) = a_0 + \cdots + a_n x^n \in F[x]$ with $a_n \neq 0$ has degree n. A root $r \in F$ has multiplicity $m \ge 1$ if $(x - r)^m \mid f(x)$ but $(x - r)^{m+1} \nmid f(x)$.

Theorem 1.12. Polynomial Division and Remainder Theorems
For $f, g \in F[x]$ with $g \neq 0$, there are unique $q, r \in F[x]$ satisfying

$$f = qg + r, \qquad r = 0 \text{ or } \deg r < \deg g.$$

In particular, the remainder on division of $f(x)$ by $x - a$ is $f(a)$, and

$$f(a) = 0 \quad \Longleftrightarrow \quad (x - a) \mid f(x).$$

Theorem 1.13. Polynomial Bézout Identity
Let F be a field and let $f, g \in F[x]$, not both zero. There are polynomials $u, v \in F[x]$ such that

$$uf + vg = \gcd(f, g),$$

where the greatest common divisor is chosen monic. In particular, f and g are coprime if and only if $uf + vg = 1$ for some $u, v \in F[x]$.

Proposition 1.14. Root-Counting Principle
A nonzero polynomial of degree d over a field has at most d distinct roots. Consequently, two polynomials of degree at most d that agree at $d + 1$ distinct points are identical.

Theorem 1.15. Viète's Formulas
Let $f(x) = a_n x^n + \cdots + a_0 = a_n \prod_{i=1}^{n} (x - r_i)$ over $\mathbb{C}$, with $a_n \neq 0$, where the roots are counted with multiplicity. Then

$$e_k(r_1, \ldots, r_n) = (-1)^k \frac{a_{n-k}}{a_n} \qquad (1 \leq k \leq n).$$

In particular, $\sum_i r_i = -a_{n-1}/a_n$ and $\prod_i r_i = (-1)^n a_0/a_n$.

Formula 1.16. Newton's Identities
Let $p_k = \sum_{i=1}^{n} r_i^k$ and let e_k be the elementary symmetric polynomials in $r_1, \ldots, r_n$. For $1 \leq k \leq n$,

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \cdots + (-1)^{k-1} e_{k-1} p_1 + (-1)^k k e_k = 0.$$

For $k > n$, the same identity holds with $e_j = 0$ for $j > n$.

Theorem 1.17. Lagrange Interpolation
Let $x_0, \ldots, x_n$ be distinct elements of a field F. There is a unique polynomial $P \in F[x]$ of degree at most n with $P(x_i) = y_i$ for all i, namely

$$P(x) = \sum_{i=0}^{n} y_i \prod_{\substack{0 \leq j \leq n \\ j \neq i}} \frac{x - x_j}{x_i - x_j}.$$

Theorem 1.18. Rational Root Theorem
Let $f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_n \neq 0$. If u/v is a root with $u, v \in \mathbb{Z}$, $v > 0$, and $\gcd(u, v) = 1$, then $u \mid a_0$ and $v \mid a_n$.

Theorem 1.19. Conjugate Root Principles
For a polynomial with real coefficients, every nonreal complex root occurs together with its complex conjugate, with the same multiplicity. For a polynomial with rational coefficients, a root of the form $a + b\sqrt{d}$ with $a, b \in \mathbb{Q}$, $b \neq 0$, and $d \in \mathbb{Q}$ not a square in $\mathbb{Q}$ occurs together with $a - b\sqrt{d}$.

Theorem 1.20. Eisenstein's Irreducibility Criterion
Let $f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ be primitive. If a prime p satisfies

$$p \mid a_0, \ldots, a_{n-1}, \qquad p \nmid a_n, \qquad p^2 \nmid a_0,$$

then f is irreducible over $\mathbb{Q}$. The criterion can also be applied after a substitution $x \mapsto x + k$ with $k \in \mathbb{Z}$.

Theorem 1.21. Cauchy's Root Bound
If $f(z) = a_n z^n + \cdots + a_0$ with $a_n \neq 0$, then every complex root z satisfies

$$|z| \leq 1 + \max_{0 \leq k < n} \left| \frac{a_k}{a_n} \right|.$$

Definition 1.22. Irreducible Polynomial and Minimal Polynomial
A nonconstant $f \in F[x]$ is *irreducible over F* if it cannot be written as a product of two polynomials of positive degree in $F[x]$. If α is algebraic over F, its *minimal polynomial* $m_{\alpha,F} \in F[x]$ is the unique monic irreducible polynomial over F satisfying $m_{\alpha,F}(\alpha) = 0$.

Theorem 1.23. Fundamental Theorem of Algebra
Every nonconstant polynomial in $\mathbb{C}[x]$ has a root in $\mathbb{C}$. Equivalently, every polynomial of degree $n \geq 1$ over $\mathbb{C}$ factors into n linear factors, counted with multiplicity.

Theorem 1.24. Combinatorial Nullstellensatz, Coefficient Form
Let F be a field and let $P \in F[x_1, \ldots, x_n]$ have total degree $t_1 + \cdots + t_n$, where $t_i \in \mathbb{N}_0$. If the coefficient of $x_1^{t_1} \cdots x_n^{t_n}$ is nonzero, then for any sets $S_i \subseteq F$ with $|S_i| > t_i$, there exist $s_i \in S_i$ such that $P(s_1, \ldots, s_n) \neq 0$.

Theorem 1.25. Gauss's Lemma for Polynomials
If $f, g \in \mathbb{Z}[x]$ are primitive polynomials, then fg is primitive. In particular, a primitive polynomial in $\mathbb{Z}[x]$ is irreducible over $\mathbb{Q}$ if and only if it is irreducible over $\mathbb{Z}$.

Theorem 1.26. Irreducibility by Reduction Modulo a Prime
Let $f \in \mathbb{Z}[x]$ have positive degree and leading coefficient not divisible by a prime p. If its reduction $\overline{f} \in \mathbb{F}_p[x]$ has the same degree and is irreducible over $\mathbb{F}_p$, then f is irreducible over $\mathbb{Q}$ provided f is primitive.

Proposition 1.27. Derivative and Multiplicity of a Root
Let $f \in F[x]$, where F has characteristic 0. An element $a \in F$ is a root of multiplicity exactly m if and only if

$$f(a) = f'(a) = \cdots = f^{(m-1)}(a) = 0, \qquad f^{(m)}(a) \neq 0.$$

Theorem 1.28. Arithmetic–Geometric Mean Inequality
For $x_1, \ldots, x_n \geq 0$,

$$\frac{x_1 + \cdots + x_n}{n} \geq (x_1 \cdots x_n)^{1/n}.$$

Equality holds exactly when $x_1 = \cdots = x_n$. More generally, for positive weights w_i with $\sum_i w_i = 1$,

$$\sum_{i=1}^{n} w_i x_i \geq \prod_{i=1}^{n} x_i^{w_i},$$

with equality when all $x_i > 0$ are equal.

Theorem 1.29. Cauchy–Schwarz Inequality
For real a_i, b_i,

$$\left(\sum_{i=1}^{n} a_i b_i \right)^2 \leq \left(\sum_{i=1}^{n} a_i^2 \right) \left(\sum_{i=1}^{n} b_i^2 \right).$$

Equality holds exactly when the two vectors are linearly dependent. For $x_i > 0$,

$$\sum_{i=1}^{n} \frac{a_i^2}{x_i} \geq \frac{(\sum_{i=1}^{n} a_i)^2}{\sum_{i=1}^{n} x_i}.$$

Theorem 1.30. Cauchy–Schwarz Inequality, Engel Form
For $x_1, \ldots, x_n \in \mathbb{R}$ and $a_1, \ldots, a_n > 0$,

$$\sum_{i=1}^{n} \frac{x_i^2}{a_i} \geq \frac{(x_1 + \cdots + x_n)^2}{a_1 + \cdots + a_n}.$$

Equality holds if and only if the ratios x_i/a_i are all equal.

Theorem 1.31. Tangent-Line Inequality for Convex Functions
If f is differentiable and convex on an open interval I, then for all $x, y \in I$,

$$f(y) \geq f(x) + f'(x)(y - x).$$

If f is strictly convex, equality holds only for $x = y$.

Theorem 1.32. Harmonic–Geometric–Arithmetic–Quadratic Means
For positive $x_1, \ldots, x_n$,

$$\frac{n}{\sum_i 1/x_i} \leq \left(\prod_i x_i\right)^{1/n} \leq \frac{\sum_i x_i}{n} \leq \sqrt{\frac{\sum_i x_i^2}{n}}.$$

Equality throughout holds exactly when all x_i are equal.

Theorem 1.33. Hölder's Inequality
Let $p, q > 1$ satisfy $1/p + 1/q = 1$. For nonnegative a_i, b_i,

$$\sum_{i=1}^{n} a_i b_i \leq \left(\sum_{i=1}^{n} a_i^p\right)^{1/p} \left(\sum_{i=1}^{n} b_i^q\right)^{1/q}.$$

For nonzero sequences, equality holds exactly when (a_i^p) and (b_i^q) are proportional.

Theorem 1.34. Minkowski's Inequality
For $p \geq 1$ and real (or complex) sequences $(a_i), (b_i)$,

$$\left(\sum_i |a_i + b_i|^p\right)^{1/p} \leq \left(\sum_i |a_i|^p\right)^{1/p} + \left(\sum_i |b_i|^p\right)^{1/p}.$$

Theorem 1.35. Jensen's Inequality
Let f be convex on an interval I, and let $x_i \in I$, $w_i \geq 0$, $\sum_i w_i = 1$. Then

$$f\left(\sum_i w_i x_i\right) \leq \sum_i w_i f(x_i).$$

For a strictly convex f and positive weights, equality holds exactly when all x_i are equal.

Theorem 1.36. Bernoulli's Inequality
For $x \geq -1$ and real $r \geq 1$,

$$(1 + x)^r \geq 1 + rx.$$

For $0 \leq r \leq 1$ the inequality reverses when $x > -1$.

Theorem 1.37. Rearrangement Inequality
If $a_1 \le \cdots \le a_n$ and $b_1 \le \cdots \le b_n$ are real, then for any permutation $\sigma \in S_n$,

$$\sum_{i=1}^{n} a_i b_{n+1-i} \le \sum_{i=1}^{n} a_i b_{\sigma(i)} \le \sum_{i=1}^{n} a_i b_i.$$

Theorem 1.38. Chebyshev's Sum Inequality
If (a_i) and (b_i) are similarly sorted real sequences, then

$$\frac{1}{n}\sum_{i=1}^{n} a_i b_i \ge \left(\frac{1}{n}\sum_{i=1}^{n} a_i\right)\left(\frac{1}{n}\sum_{i=1}^{n} b_i\right).$$

The inequality reverses if the two sequences are oppositely sorted.

Theorem 1.39. Schur's Inequality
For $a, b, c > 0$ and $r \ge 0$,

$$\sum_{\text{cyc}} a^r (a-b)(a-c) \ge 0.$$

For $r = 1$, this becomes

$$a(a-b)(a-c) + b(b-c)(b-a) + c(c-a)(c-b) \ge 0.$$

Theorem 1.40. Nesbitt's Inequality
For $a, b, c > 0$,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \ge \frac{3}{2}.$$

Equality holds exactly when $a = b = c$.

Theorem 1.41. Young's Inequality for Products
For $a, b \ge 0$ and $p, q > 1$ with $1/p + 1/q = 1$,

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}.$$

Equality holds exactly when $a^p = b^q$.

Definition 1.42. Majorization
For vectors $x, y \in \mathbb{R}^n$, let $x_{[1]} \ge \cdots \ge x_{[n]}$ and $y_{[1]} \ge \cdots \ge y_{[n]}$ be their coordinates in decreasing order. We say that x *majorizes* y if

$$\sum_{i=1}^{k} x_{[i]} \ge \sum_{i=1}^{k} y_{[i]} \quad (1 \le k < n), \qquad \sum_i x_i = \sum_i y_i.$$

Theorem 1.43. Muirhead's Inequality
Let $a_1, \ldots, a_n \ge 0$. For a nonnegative integer n-tuple α, let $m_\alpha(a)$ denote the average of $a_1^{\alpha_{\sigma(1)}} \cdots a_n^{\alpha_{\sigma(n)}}$ over all $\sigma \in S_n$. If α majorizes β, then

$$m_\alpha(a) \ge m_\beta(a).$$

Theorem 1.44. Power Mean Inequality
For positive numbers $x_1, \dots, x_n$ and real r, let

$$M_r(x) = \begin{cases} (\frac{1}{n}\sum_i x_i^r)^{1/r}, & r \neq 0, \\ (\prod_i x_i)^{1/n}, & r = 0. \end{cases}$$

If $r < s$, then $M_r(x) \leq M_s(x)$, with equality exactly when all x_i are equal.

Theorem 1.45. Maclaurin's Inequalities
For positive $x_1, \dots, x_n$, let e_k be the elementary symmetric sums and put

$$E_k = \frac{e_k}{\binom{n}{k}} \qquad (1 \leq k \leq n).$$

Then

$$E_1 \geq E_2^{1/2} \geq \cdots \geq E_n^{1/n},$$

with equality throughout exactly when $x_1 = \cdots = x_n$.

Theorem 1.46. Karamata's Inequality
Let $x, y \in I^n$ for an interval I, and suppose x majorizes y. For a convex function $f\colon I \to \mathbb{R}$,

$$\sum_{i=1}^{n} f(x_i) \geq \sum_{i=1}^{n} f(y_i).$$

If f is strictly convex, equality holds exactly when x is a permutation of y.

Definition 1.47. Sequence and Linear Recurrence
A sequence (a_n) over a field F is a function from an index set such as $\mathbb{N}_0$ to F. It satisfies a linear recurrence of order k with constant coefficients if

$$a_{n+k} = c_1 a_{n+k-1} + \cdots + c_k a_n$$

for fixed $c_1, \dots, c_k \in F$, with $c_k \neq 0$.

Formula 1.48. Arithmetic and Geometric Progressions
For $n \geq 1$, an arithmetic progression $a_n = a_1 + (n-1)d$ satisfies

$$\sum_{k=1}^{n} a_k = \frac{n(a_1 + a_n)}{2}.$$

A geometric progression $a_n = a_1 r^{n-1}$ satisfies

$$\sum_{k=1}^{n} a_1 r^{k-1} = \begin{cases} a_1(1-r^n)/(1-r), & r \neq 1, \\ n a_1, & r = 1. \end{cases}$$

Theorem 1.49. Linear Recurrences and Characteristic Roots
Suppose a sequence over $\mathbb{C}$ satisfies

$$a_{n+k} = c_1 a_{n+k-1} + \cdots + c_k a_n$$

with $c_k \neq 0$. If the characteristic polynomial

$$P(t) = t^k - c_1 t^{k-1} - \cdots - c_k$$

has distinct roots r_i of multiplicities m_i, then every solution is of the form

$$a_n = \sum_i Q_i(n) r_i^n, \qquad \deg Q_i < m_i.$$

The constants in the polynomials Q_i are determined uniquely by k consecutive initial values.

Formula 1.50. Fibonacci Numbers and Binet's Formula
Let $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. Then

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \quad \psi = \frac{1-\sqrt{5}}{2}.$$

For $m, n \geq 1$,

$$F_{m+n} = F_m F_{n+1} + F_{m-1} F_n, \qquad F_{n+1}F_{n-1} - F_n^2 = (-1)^n.$$

Formula 1.51. Telescoping Identities
For any sequence (a_k) and $m \leq n$,

$$\sum_{k=m}^{n} (a_{k+1} - a_k) = a_{n+1} - a_m.$$

If $a_k, a_{k+1} \neq 0$,

$$\frac{a_{k+1} - a_k}{a_k a_{k+1}} = \frac{1}{a_k} - \frac{1}{a_{k+1}}.$$

Formula 1.52. Finite Differences and Newton's Formula
For a sequence $(a_n)_{n\geq 0}$, define $(\Delta a)_n = a_{n+1} - a_n$. If $\Delta^{d+1} a = 0$, then

$$a_n = \sum_{k=0}^{d} \binom{n}{k} (\Delta^k a)_0 \qquad (n \in \mathbb{N}_0).$$

In particular, if $a_n = P(n)$ for a polynomial P of degree at most d, then $\Delta^{d+1} a = 0$.

Formula 1.53. First-Order Linear Recurrence
If $a_{n+1} = r a_n + b_n$ for $n \geq 0$, with fixed $r \in \mathbb{C}$, then

$$a_n = r^n a_0 + \sum_{j=0}^{n-1} r^{n-1-j} b_j \qquad (n \geq 1).$$

If $b_n = b$ is constant and $r \neq 1$, this becomes

$$a_n = r^n a_0 + b\frac{r^n - 1}{r - 1}.$$

Definition 1.54. Injective, Surjective, and Bijective Functions
A function $f\colon X \to Y$ is injective if $f(x) = f(y)$ implies $x = y$, surjective if every $y \in Y$ equals $f(x)$ for some $x \in X$, and bijective if both properties hold. A bijection has an inverse function $f^{-1}\colon Y \to X$.

Definition 1.55. Additive and Multiplicative Functions
A function $f\colon \mathbb{R} \to \mathbb{R}$ is additive if $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. A function $g\colon \mathbb{R} \to \mathbb{R}$ is multiplicative if $g(xy) = g(x)g(y)$ for all $x, y \in \mathbb{R}$. These notions are distinct from multiplicative arithmetic functions on $\mathbb{N}$.

Theorem 1.56. Cauchy's Additive Functional Equation
If $f\colon \mathbb{R} \to \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$, then $f(q) = qf(1)$ for all $q \in \mathbb{Q}$. If, in addition, f is continuous at one point, monotone on an interval, Lebesgue measurable, or bounded on an interval of positive length, then

$$f(x) = cx \qquad (x \in \mathbb{R})$$

for some $c \in \mathbb{R}$.

Theorem 1.57. Jensen's Functional Equation
If $f\colon \mathbb{R} \to \mathbb{R}$ satisfies

$$f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \qquad (x, y \in \mathbb{R}),$$

then $f(x) = A(x)+b$ for an additive function $A\colon \mathbb{R} \to \mathbb{R}$ and a constant $b \in \mathbb{R}$. If f is continuous at one point, then $f(x) = cx + b$ for real b, c.

Theorem 1.58. Continuous Exponential Equation
If a continuous function $f\colon \mathbb{R} \to (0, \infty)$ satisfies

$$f(x+y) = f(x)f(y) \qquad (x, y \in \mathbb{R}),$$

then $f(x) = e^{cx}$ for some $c \in \mathbb{R}$.

Remark. In a functional equation, the domain, codomain, and any regularity assumptions are part of the statement. The same formal equation may have additional nonstandard solutions without continuity, measurability, or related hypotheses.

Theorem 1.59. Polynomial Functional Equations
Suppose $P, Q \in F[x]$ over an infinite field F satisfy $P(t) = Q(t)$ for every $t \in F$. Then $P = Q$ as polynomials, so their corresponding coefficients are equal.

Formula 1.60. Polar Form and De Moivre's Formula
For $z = r(\cos\theta + i\sin\theta)$ with $r \geq 0$ and $n \in \mathbb{Z}$ (when $n < 0$, assume $r > 0$),

$$z^n = r^n(\cos(n\theta) + i\sin(n\theta)).$$

The nth roots of a nonzero complex number $re^{i\theta}$ are

$$r^{1/n}e^{i(\theta+2\pi k)/n}, \qquad k = 0, \ldots, n-1.$$

Definition 1.61. Roots of Unity
For $n \geq 1$, the nth roots of unity are the solutions of $z^n = 1$. They form a cyclic group of order n, generated by

$$\zeta_n = e^{2\pi i/n}.$$

An nth root of unity is *primitive* if its multiplicative order is n.

Formula 1.62. Sums of Roots of Unity
For $n \geq 2$ and any integer m,

$$\sum_{k=0}^{n-1} \zeta_n^{mk} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

The same identity holds for any primitive nth root in place of ζ_n.

Formula 1.63. Roots-of-Unity Filter
If $P(x) = \sum_{k=0}^{N} a_k x^k \in \mathbb{C}[x]$ and $r \in \mathbb{Z}$, then

$$\sum_{\substack{0 \leq k \leq N \\ k \equiv r \pmod{n}}} a_k = \frac{1}{n}\sum_{j=0}^{n-1} \zeta_n^{-rj} P(\zeta_n^j).$$

Theorem 1.64. Cyclotomic Factorization
For $n \geq 1$, let $\Phi_n(x)$ be the monic polynomial whose roots are the primitive nth roots of unity. Then $\Phi_n(x) \in \mathbb{Z}[x]$, $\deg \Phi_n = \varphi(n)$, and

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x).$$

Each Φ_n is irreducible over $\mathbb{Q}$. Here $\varphi(n)$ denotes the number of integers $1 \leq k \leq n$ relatively prime to n.

Definition 1.65. Group and Subgroup
A group $(G, *)$ is a set with an associative binary operation, an identity e, and a two-sided inverse for each element. A subset $H \subseteq G$ is a subgroup, written $H \leq G$, if it is a group under the restricted operation. The group is *abelian* if $a * b = b * a$ for all $a, b \in G$.

Definition 1.66. Cyclic and Permutation Groups
A group generated by one element g is cyclic, written $\langle g \rangle$. Write C_n for the cyclic group of order n and S_n for the permutation group on n symbols. The dihedral group D_n is the symmetry group of a regular n-gon and has order $2n$.

Theorem 1.67. Lagrange's Theorem
If $H \leq G$ and G is finite, then

$$|G| = [G : H]|H|.$$

Consequently, $|H| \mid |G|$ and the order $\operatorname{ord}(g)$ of any $g \in G$ divides $|G|$; in particular, $g^{|G|} = e$.

Theorem 1.68. Cauchy's Theorem for Finite Groups
If a prime p divides the order of a finite group G, then G has an element of order p.

Theorem 1.69. Structure of Finite Cyclic Groups
If $G = \langle g \rangle$ has order n, then

$$\operatorname{ord}(g^k) = \frac{n}{\gcd(n, k)}.$$

For each positive divisor $d \mid n$ there is exactly one subgroup of G of order d, and G has exactly $\varphi(n)$ generators, where φ is Euler's totient function.

Definition 1.70. Homomorphism, Kernel, and Quotient Group
A group homomorphism $\varphi \colon G \to H$ satisfies $\varphi(xy) = \varphi(x)\varphi(y)$. Its kernel is $\ker \varphi = \{g \in G \mid \varphi(g) = e_H\}$. A subgroup $N \leq G$ is normal if $gNg^{-1} = N$ for all $g \in G$; in that case G/N is the quotient group of cosets.

Theorem 1.71. First Isomorphism Theorem for Groups
For any group homomorphism $\varphi \colon G \to H$,

$$G/\ker \varphi \cong \operatorname{im} \varphi.$$

Definition 1.72. Group Action, Orbit, and Stabilizer
A left action of G on X is a map $(g, x) \mapsto g \cdot x$ satisfying $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$. The orbit of x is $G \cdot x = \{g \cdot x \mid g \in G\}$, and the stabilizer is $\operatorname{Stab}_G(x) = \{g \in G \mid g \cdot x = x\}$.

Theorem 1.73. Orbit–Stabilizer Theorem
If a finite group G acts on a set X, then for each $x \in X$,

$$|G \cdot x| = [G : \operatorname{Stab}_G(x)] = \frac{|G|}{|\operatorname{Stab}_G(x)|}.$$

Theorem 1.74. Burnside's Lemma; Cauchy–Frobenius Formula
If a finite group G acts on a finite set X, the number of orbits is

$$|G\backslash X| = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|,$$

where $\operatorname{Fix}(g) = \{x \in X \mid g \cdot x = x\}$.

Theorem 1.75. Class Equation
For a finite group G, choose representatives $x_1, \ldots, x_r$ of the conjugacy classes not contained in its center $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(x_i)].$$

Theorem 1.76. Sylow Theorems
Let $|G| = p^k m$ where p is prime and $p \nmid m$. Then G has a subgroup of order p^k, all such subgroups are conjugate, and their number n_p satisfies

$$n_p \equiv 1 \pmod{p}, \qquad n_p \mid m.$$

Theorem 1.77. Fundamental Theorem of Finite Abelian Groups
Every finite abelian group is isomorphic to a direct product of cyclic groups of prime-power order. Equivalently, it is isomorphic to

$$C_{d_1} \times \cdots \times C_{d_r}, \qquad d_1 \mid d_2 \mid \cdots \mid d_r,$$

with the invariant factors $d_i > 1$ uniquely determined (the trivial group corresponds to an empty product).

Theorem 1.78. Sign and Cycle Decomposition
Every permutation $\sigma \in S_n$ is a product of disjoint cycles, uniquely up to their ordering and cyclic rotation. If its disjoint-cycle decomposition has c cycles, including fixed points, then

$$\operatorname{sgn}(\sigma) = (-1)^{n-c}.$$

Definition 1.79. Ring, Integral Domain, and Field
Unless stated otherwise, rings are associative with an identity and need not be commutative. An integral domain is a nonzero commutative ring with identity and without zero divisors. A field is a nonzero commutative ring in which every nonzero element is invertible.

Definition 1.80. Ideal, Quotient Ring, and Characteristic
An ideal I of a ring R means a two-sided ideal. The quotient R/I is a ring. The characteristic $\operatorname{char} R$ is the additive order of 1_R, or 0 if it has infinite additive order. Ring homomorphisms preserve identities.

Theorem 1.81. Finite Integral Domains
Every finite integral domain is a field. A finite field has cardinality p^k for some prime p and $k \geq 1$, and for each such prime power q there is, up to isomorphism, a unique field $\mathbb{F}_q$.

Theorem 1.82. Lagrange's Theorem on Finite Subgroups of a Field
Every finite subgroup G of the multiplicative group $F^\times$ of a field F is cyclic. In particular, the multiplicative group of each finite field is cyclic.

Theorem 1.83. Multiplicative Group of a Finite Field
The group $\mathbb{F}_q^\times$ is cyclic of order $q-1$. In particular, each $a \in \mathbb{F}_q$ satisfies $a^q = a$, and for $m \in \mathbb{N}$ the equation $x^m = 1$ in $\mathbb{F}_q$ has exactly $\gcd(m, q-1)$ solutions.

Proposition 1.84. Irreducible Polynomial Quotients
For a field F and nonconstant $f \in F[x]$, the quotient ring $F[x]/(f)$ is a field if and only if f is irreducible over F. If $F = \mathbb{F}_q$ and $\deg f = d$, the quotient has q^d elements.

Theorem 1.85. Polynomial Method over Finite Fields
For a nonzero polynomial $f \in \mathbb{F}_q[x]$ of degree d, the equation $f(x) = 0$ has at most d solutions in $\mathbb{F}_q$. A polynomial function on $\mathbb{F}_q$ has a unique representative of degree at most $q-1$ modulo $x^q - x$.

Chapter 2

Number Theory

Definition 2.1. Divisibility, Primes, and Coprimality
For $a, b \in \mathbb{Z}$, write $a \mid b$ if $b = ak$ for some $k \in \mathbb{Z}$. A prime is an integer $p > 1$ whose only positive divisors are 1 and p. Two integers a, b, not both zero, are coprime if $\gcd(a, b) = 1$.

Theorem 2.2. Division Algorithm
For $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, there exist unique $q, r \in \mathbb{Z}$ such that

$$a = bq + r, \qquad 0 \leq r < b.$$

Proposition 2.3. Euclidean Algorithm
For integers a, b with $b \neq 0$,

$$\gcd(a, b) = \gcd(b, a \bmod |b|).$$

Repeated Euclidean division produces the greatest common divisor as the last nonzero remainder.

Theorem 2.4. Fundamental Theorem of Arithmetic
Every integer $n > 1$ has a factorization

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k},$$

where the p_i are distinct primes and $\alpha_i \in \mathbb{N}$. It is unique up to the order of the prime factors.

Theorem 2.5. Euclid's Lemma
If a prime p divides ab, then $p \mid a$ or $p \mid b$. More generally, if $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

Theorem 2.6. Bézout's Identity
For integers a, b, not both zero, there exist $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b).$$

Moreover, the set of integer linear combinations $ax + by$ equals the set of multiples of $\gcd(a, b)$.

Formula 2.7. GCD and LCM from Prime Factorizations
For positive integers a, b and prime-power exponents $a = \prod_p p^{\alpha_p}$, $b = \prod_p p^{\beta_p}$ (allowing zero exponents),

$$\gcd(a, b) = \prod_p p^{\min(\alpha_p, \beta_p)}, \qquad \operatorname{lcm}(a, b) = \prod_p p^{\max(\alpha_p, \beta_p)}.$$

Consequently,

$$\gcd(a, b) \operatorname{lcm}(a, b) = ab.$$

Theorem 2.8. GCD of Differences of Powers
For integers $a > 1$ and $m, n \in \mathbb{N}$,

$$\gcd(a^m - 1, a^n - 1) = a^{\gcd(m,n)} - 1.$$

In particular, if $m \mid n$, then $a^m - 1 \mid a^n - 1$.

Theorem 2.9. Euclid's Theorem
There are infinitely many prime numbers.

Theorem 2.10. Bertrand's Postulate; Chebyshev's Theorem
For every integer $n > 1$, there is a prime p satisfying

$$n < p < 2n.$$

Definition 2.11. Congruence and Residue Class
For $n \in \mathbb{N}$,

$$a \equiv b \pmod{n} \quad \iff \quad n \mid (a - b).$$

The residue class of a modulo n is $[a]_n \in \mathbb{Z}/n\mathbb{Z}$. The group $(\mathbb{Z}/n\mathbb{Z})^\times$ consists of the classes $[a]_n$ with $\gcd(a, n) = 1$.

Proposition 2.12. Algebra of Congruences
If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d \pmod{n}, \qquad ac \equiv bd \pmod{n}.$$

If $ac \equiv bc \pmod{n}$ and $g = \gcd(c, n)$, then $a \equiv b \pmod{n/g}$; cancellation modulo n is valid when $\gcd(c, n) = 1$.

Theorem 2.13. Linear Congruence Criterion
The congruence $ax \equiv b \pmod{n}$, with $n \geq 1$, has an integer solution if and only if $d = \gcd(a, n)$ divides b. In that case there are exactly d incongruent solutions modulo n.

Theorem 2.14. Chinese Remainder Theorem
If $m_1, \ldots, m_k \in \mathbb{N}$ are pairwise coprime, then for arbitrary integers $a_1, \ldots, a_k$, the system

$$x \equiv a_i \pmod{m_i} \qquad (1 \leq i \leq k)$$

has a unique solution modulo $M = m_1 \cdots m_k$.

Proposition 2.15. Chinese Remainder Theorem for General Moduli
For positive moduli $m_1, \ldots, m_k$, the congruences $x \equiv a_i \pmod{m_i}$ are simultaneously solvable if and only if

$$a_i \equiv a_j \pmod{\gcd(m_i, m_j)} \qquad \text{for all } i, j.$$

When solvable, the solution is unique modulo $\operatorname{lcm}(m_1, \ldots, m_k)$.

Formula 2.16. Decimal Divisibility Tests
For decimal digits $d_0, \ldots, d_m$ and $N = \sum_{k=0}^{m} d_k 10^k$,

$$N \equiv \sum_{k=0}^{m} d_k \pmod{9}, \qquad N \equiv \sum_{k=0}^{m} (-1)^k d_k \pmod{11}.$$

The last r digits determine $N \pmod{2^r}$ and $N \pmod{5^r}$.

Formula 2.17. Polynomial Congruences
If $f \in \mathbb{Z}[x]$ and $a \equiv b \pmod{n}$, then $f(a) \equiv f(b) \pmod{n}$. In particular, the integral roots of $f(x) = 0$ satisfy every reduction of that equation modulo n.

Theorem 2.18. Wilson's Theorem
An integer $n > 1$ is prime if and only if

$$(n-1)! \equiv -1 \pmod{n}.$$

Proposition 2.19. Units and Inverses Modulo n
An integer a has a multiplicative inverse modulo n if and only if $\gcd(a, n) = 1$. *If integers u, v satisfy $au + nv = 1$, then u represents an inverse of a modulo n.*

Definition 2.20. Arithmetic and Multiplicative Functions
An arithmetic function is a map $f\colon \mathbb{N} \to \mathbb{C}$. It is multiplicative if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. It is completely multiplicative if this identity holds for all $m, n \in \mathbb{N}$.

Definition 2.21. Euler's Totient Function
The Euler totient function is

$$\varphi(n) = |\{1 \le k \le n \mid \gcd(k, n) = 1\}|.$$

It equals the order of the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$.

Formula 2.22. Euler's Totient Formula
For $n = \prod_{i=1}^{r} p_i^{\alpha_i}$ with $\alpha_i \ge 1$,

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right) = \prod_{i=1}^{r} p_i^{\alpha_i - 1}(p_i - 1).$$

The empty product gives $\varphi(1) = 1$. Also

$$\sum_{d \mid n} \varphi(d) = n.$$

Formula 2.23. Divisor Functions
For $n = \prod_{i=1}^{r} p_i^{\alpha_i}$, the number $\tau(n)$ and sum $\sigma(n)$ of positive divisors satisfy

$$\tau(n) = \prod_{i=1}^{r} (\alpha_i + 1), \qquad \sigma(n) = \prod_{i=1}^{r} \frac{p_i^{\alpha_i + 1} - 1}{p_i - 1}.$$

More generally, $\sigma_k(n) = \sum_{d \mid n} d^k = \prod_i (1 + p_i^k + \cdots + p_i^{k\alpha_i})$ for $k \in \mathbb{N}_0$.

Definition 2.24. Möbius Function
The Möbius function $\mu\colon \mathbb{N} \to \{-1, 0, 1\}$ is given by

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n \text{ is a product of } r \text{ distinct primes}, \\ 0, & n \text{ is divisible by a square greater than 1}. \end{cases}$$

Theorem 2.25. Möbius Inversion Formula
For arithmetic functions f, g, if

$$g(n) = \sum_{d|n} f(d),$$

then

$$f(n) = \sum_{d|n} \mu(d)g(n/d) = \sum_{d|n} \mu(n/d)g(d).$$

In particular,

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Definition 2.26. Dirichlet Convolution
For arithmetic functions f, g, their Dirichlet convolution is

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d).$$

It is associative and commutative; the identity is $\varepsilon(1) = 1$, $\varepsilon(n) = 0$ for $n > 1$. The constant function $\mathbf{1}(n) = 1$ satisfies $\mu * \mathbf{1} = \varepsilon$.

Proposition 2.27. Multiplicative Arithmetic Functions
If f, g are multiplicative arithmetic functions, then their Dirichlet convolution $f * g$ is multiplicative. Moreover, τ, σ_k, φ, and μ are multiplicative.

Theorem 2.28. Fermat's Little Theorem
If p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, $a^p \equiv a \pmod{p}$ for every integer a.

Theorem 2.29. Euler's Theorem
For $n \in \mathbb{N}$ and $\gcd(a, n) = 1$,

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Definition 2.30. Multiplicative Order
For $\gcd(a, n) = 1$, the multiplicative order $\operatorname{ord}_n(a)$ is the least positive integer r such that $a^r \equiv 1 \pmod{n}$. It divides $\varphi(n)$, and

$$a^k \equiv 1 \pmod{n} \iff \operatorname{ord}_n(a) \mid k$$

for $k \in \mathbb{Z}$.

Definition 2.31. Primitive Root
An integer g is a primitive root modulo n if $\gcd(g, n) = 1$ and

$$\operatorname{ord}_n(g) = \varphi(n).$$

Equivalently, $[g]_n$ generates $(\mathbb{Z}/n\mathbb{Z})^\times$.

Theorem 2.32. Primitive Root Theorem
For $n > 1$, primitive roots modulo n exist exactly when

$$n \in \{2, 4, p^k, 2p^k\},$$

where p is an odd prime and $k \in \mathbb{N}$.

Proposition 2.33. Power Congruences in a Cyclic Group
Suppose $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic of order $N = \varphi(n)$. For an integer $k \geq 1$ and $\gcd(a, n) = 1$, the congruence

$$x^k \equiv a \pmod{n}$$

has solutions if and only if $a^{N/d} \equiv 1 \pmod{n}$, where $d = \gcd(k, N)$. When solvable, it has exactly d solutions modulo n.

Definition 2.34. Carmichael Function
For $n \geq 1$, the Carmichael function $\lambda(n)$ is the exponent of $(\mathbb{Z}/n\mathbb{Z})^\times$: the least positive integer m such that $a^m \equiv 1 \pmod{n}$ for all a coprime to n. For odd primes p,

$$\lambda(p^k) = p^{k-1}(p-1),$$

while $\lambda(2) = 1$, $\lambda(4) = 2$, and $\lambda(2^k) = 2^{k-2}$ for $k \geq 3$. For pairwise coprime a, b, $\lambda(ab) = \operatorname{lcm}(\lambda(a), \lambda(b))$.

Theorem 2.35. Primitive Root Test
Let p be an odd prime and $g \in \mathbb{Z}$ with $p \nmid g$. Then g is a primitive root modulo p if and only if

$$g^{(p-1)/\ell} \not\equiv 1 \pmod{p}$$

for every distinct prime divisor ℓ of $p - 1$.

Proposition 2.36. The Group of Units Modulo a Power of Two
For $k \geq 3$,

$$(\mathbb{Z}/2^k\mathbb{Z})^\times \cong C_2 \times C_{2^{k-2}}.$$

In particular, there is no primitive root modulo 2^k when $k \geq 3$.

Definition 2.37. p-Adic Valuation
For a prime p and nonzero integer n, $v_p(n)$ is the largest $r \in \mathbb{N}_0$ such that $p^r \mid n$. For nonzero integers a, b,

$$v_p(ab) = v_p(a) + v_p(b), \qquad v_p(a+b) \geq \min(v_p(a), v_p(b))$$

when $a + b \neq 0$. The second inequality is an equality if $v_p(a) \neq v_p(b)$.

Theorem 2.38. Legendre's Formula
For a prime p and $n \in \mathbb{N}$,

$$v_p(n!) = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor = \frac{n - s_p(n)}{p-1},$$

where $s_p(n)$ is the sum of the base-p digits of n.

Theorem 2.39. Lifting the Exponent Lemma; LTE
Let p be an odd prime, $n \in \mathbb{N}$, and $x, y \in \mathbb{Z}$ with $x \neq y$, $p \mid x - y$, and $p \nmid xy$. Then

$$v_p(x^n - y^n) = v_p(x - y) + v_p(n).$$

If instead $p \mid x + y$, $p \nmid xy$, n is odd, and $x^n + y^n \neq 0$, then

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n).$$

For distinct odd integers x, y and even n,

$$v_2(x^n - y^n) = v_2(x - y) + v_2(x + y) + v_2(n) - 1.$$

For odd n, $v_2(x^n - y^n) = v_2(x - y)$.

Theorem 2.40. Lucas's Theorem
Let p be prime and write $n = \sum_{i\geq 0} n_i p^i$, $k = \sum_{i\geq 0} k_i p^i$, with $0 \leq n_i, k_i < p$. For $0 \leq k \leq n$,

$$\binom{n}{k} \equiv \prod_{i\geq 0} \binom{n_i}{k_i} \pmod{p},$$

where $\binom{a}{b} = 0$ if $b > a$.

Theorem 2.41. Kummer's Theorem
For a prime p and integers $0 \leq k \leq n$, the valuation $v_p\left(\binom{n}{k}\right)$ equals the number of carries when k and $n-k$ are added in base p.

Formula 2.42. Valuation of a Binomial Coefficient
For a prime p and $0 \leq k \leq n$,

$$v_p\left(\binom{n}{k}\right) = \sum_{r\geq 1} \left(\left\lfloor \frac{n}{p^r} \right\rfloor - \left\lfloor \frac{k}{p^r} \right\rfloor - \left\lfloor \frac{n-k}{p^r} \right\rfloor \right).$$

Formula 2.43. Powers of Two in Factorials
For $n \in \mathbb{N}$,

$$v_2(n!) = n - s_2(n),$$

where $s_2(n)$ is the number of ones in the binary expansion of n.

Proposition 2.44. Valuations of Products
For nonzero integers $a_1, \ldots, a_m$,

$$v_p\left(\prod_{i=1}^{m} a_i\right) = \sum_{i=1}^{m} v_p(a_i).$$

For nonzero integers a, b,

$$v_p(\gcd(a,b)) = \min(v_p(a), v_p(b)),$$

where $\gcd(a,b)$ is positive.

Definition 2.45. Quadratic Residue and Legendre Symbol
For an odd prime p and $a \in \mathbb{Z}$, a is a quadratic residue modulo p if $x^2 \equiv a \pmod{p}$ for some integer x. The Legendre symbol is

$$\left(\frac{a}{p}\right) = \begin{cases} 0, & p \mid a, \\ 1, & a \text{ is a nonzero quadratic residue modulo } p, \\ -1, & \text{otherwise.} \end{cases}$$

Theorem 2.46. Euler's Criterion
For an odd prime p and any integer a,

$$a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}.$$

If $p \nmid a$, there are exactly two solutions modulo p to $x^2 \equiv a \pmod{p}$ when the Legendre symbol is 1, and none when it is -1.

Proposition 2.47. Multiplicativity of the Legendre Symbol
For an odd prime p and integers a, b,

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right).$$

Theorem 2.48. Gauss's Lemma for Quadratic Residues
Let p be an odd prime, $a \in \mathbb{Z}$ with $p \nmid a$, and let m be the number of integers $k \in \{1, \ldots, (p-1)/2\}$ for which the least positive residue of ak modulo p exceeds $p/2$. Then

$$\left(\frac{a}{p}\right) = (-1)^m.$$

Theorem 2.49. Law of Quadratic Reciprocity
For distinct odd primes p, q,

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{(p-1)(q-1)/4}.$$

The supplementary laws are

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}, \qquad \left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}.$$

Definition 2.50. Jacobi Symbol
For a positive odd integer $n = \prod_i p_i^{\alpha_i}$ and $a \in \mathbb{Z}$, define

$$\left(\frac{a}{n}\right) = \prod_i \left(\frac{a}{p_i}\right)^{\alpha_i}.$$

By convention $\left(\frac{a}{1}\right) = 1$. For composite n, Jacobi symbol 1 does not necessarily imply that a is a quadratic residue modulo n.

Proposition 2.51. Solutions of $x^2 \equiv 1$ Modulo Prime Powers
For an odd prime p and $k \geq 1$, the solutions of $x^2 \equiv 1 \pmod{p^k}$ are exactly $x \equiv \pm 1 \pmod{p^k}$. Modulo 2^k, there is one solution for $k = 1$, two for $k = 2$, and four for $k \geq 3$.

Theorem 2.52. Quadratic Residues of -1 and 2
For an odd prime p,

$$-1 \text{ is a quadratic residue modulo } p \iff p \equiv 1 \pmod 4,$$

$$2 \text{ is a quadratic residue modulo } p \iff p \equiv \pm 1 \pmod 8.$$

Definition 2.53. Diophantine Equation
A Diophantine equation is an equation in which solutions are required to be integers (or, when explicitly specified, natural numbers or rational numbers). The domain restriction is part of the problem.

Theorem 2.54. Linear Diophantine Equations
Let $a, b, c \in \mathbb{Z}$, with a, b not both zero. The equation $ax + by = c$ has integer solutions if and only if $d = \gcd(a, b)$ divides c. If (x_0, y_0) is one solution, all solutions are

$$x = x_0 + \frac{b}{d}t, \qquad y = y_0 - \frac{a}{d}t, \qquad t \in \mathbb{Z}.$$

Theorem 2.55. Primitive Pythagorean Triples
All triples of positive integers a, b, c satisfying $a^2 + b^2 = c^2$ and $\gcd(a, b, c) = 1$ arise, after possibly interchanging a, b, as

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2,$$

where $m > n \geq 1$, $\gcd(m, n) = 1$, and m, n have opposite parity.

Theorem 2.56. Frobenius Coin Theorem for Two Denominations
If $a, b \geq 2$ are coprime integers, the greatest nonnegative integer not expressible as $ax + by$ with $x, y \in \mathbb{N}_0$ is

$$ab - a - b.$$

The number of nonnegative integers not expressible in this form is $(a - 1)(b - 1)/2$.

Theorem 2.57. Fermat's Two-Square Theorem
An odd prime p can be written as a sum of two integer squares if and only if $p \equiv 1 \pmod 4$. The prime $2 = 1^2 + 1^2$ is also such a sum.

Theorem 2.58. Sum of Two Squares Criterion
A positive integer n is a sum of two integer squares if and only if every prime $p \equiv 3 \pmod 4$ appears to an even exponent in the prime factorization of n.

Formula 2.59. Brahmagupta–Fibonacci Identity
For integers a, b, c, d,

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2.$$

The companion expression $(ac + bd)^2 + (ad - bc)^2$ has the same value.

Theorem 2.60. Lagrange's Four-Square Theorem
Every nonnegative integer is a sum of four integer squares.

Proposition 2.61. Solutions of Homogeneous Linear Diophantine Equations
If $\gcd(a, b) = d$ and a, b are not both zero, then every integer solution of $ax + by = 0$ has the form

$$(x, y) = \left(\frac{b}{d}t, -\frac{a}{d}t\right), \qquad t \in \mathbb{Z}.$$

Theorem 2.62. Zsigmondy's Theorem, Integer Form
Let $a > b \geq 1$ be coprime integers and $n > 1$. Except when $n = 2$ and $a + b$ is a power of 2, or when $(a, b, n) = (2, 1, 6)$, the integer $a^n - b^n$ has a prime divisor which divides no $a^k - b^k$ with $1 \leq k < n$.

Definition 2.63. Simple Continued Fraction
For integers $a_0 \in \mathbb{Z}$, $a_i \in \mathbb{N}$ for $i \geq 1$, the simple continued fraction

$$[a_0; a_1, \ldots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cdots + \cfrac{1}{a_n}}}$$

has a reduced form p_n/q_n with $q_n > 0$, called its nth convergent.

Proposition 2.64. Convergent Recurrences
With initial values $p_{-2} = 0$, $p_{-1} = 1$, $q_{-2} = 1$, $q_{-1} = 0$, the convergents satisfy

$$p_n = a_n p_{n-1} + p_{n-2}, \qquad q_n = a_n q_{n-1} + q_{n-2},$$

and

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}.$$

Theorem 2.65. Approximation by Continued Fractions
If $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ has convergents p_n/q_n, then

$$\left| \alpha - \frac{p_n}{q_n} \right| < \frac{1}{q_n^2}.$$

Definition 2.66. Pell Equation
For a fixed positive nonsquare integer D, Pell's equation is

$$x^2 - Dy^2 = 1, \qquad x, y \in \mathbb{Z}.$$

Theorem 2.67. Solutions of Pell's Equation
For every positive nonsquare integer D, there is a least solution (x_1, y_1) in positive integers to $x^2 - Dy^2 = 1$ with $x_1 > 1$. Every positive-integer solution (x_n, y_n) is determined by

$$x_n + y_n \sqrt{D} = (x_1 + y_1 \sqrt{D})^n, \qquad n \in \mathbb{N}.$$

Theorem 2.68. Periodicity of Continued Fractions of Quadratic Irrationals
A real irrational number has an eventually periodic simple continued fraction if and only if it is a quadratic irrational, that is, a root of an irreducible quadratic polynomial over $\mathbb{Q}$.

Definition 2.69. Gaussian Integers and Norm
The Gaussian integers form the ring $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$. The norm is

$$N(a + bi) = a^2 + b^2 = (a + bi)(a - bi).$$

It is multiplicative: $N(zw) = N(z)N(w)$. The units of $\mathbb{Z}[i]$ are $\pm 1, \pm i$.

Theorem 2.70. Unique Factorization in Gaussian Integers
The ring $\mathbb{Z}[i]$ is a Euclidean domain with respect to its norm and hence a unique factorization domain. Every nonzero nonunit factors into Gaussian primes, uniquely up to order and multiplication by units.

Theorem 2.71. Finite Fields and Their Multiplicative Groups
For each prime power q, there is up to isomorphism a unique field $\mathbb{F}_q$ with q elements. Its nonzero elements form a cyclic group of order $q - 1$, so every $a \in \mathbb{F}_q$ satisfies $a^q = a$.

Theorem 2.72. Hensel's Lemma, Simple-Root Form
Let p be prime and $f \in \mathbb{Z}[x]$. If an integer a satisfies

$$f(a) \equiv 0 \pmod{p}, \qquad f'(a) \not\equiv 0 \pmod{p},$$

then for every $k \in \mathbb{N}$ there exists a unique residue class $x_k \pmod{p^k}$ with $x_k \equiv a \pmod{p}$ and $f(x_k) \equiv 0 \pmod{p^k}$.

Chapter 3

Geometry

Definition 3.1. Segments, Angles, and Directed Angles
For points A, B, write $\overline{AB}$ for the segment and AB for its length. The angle $\angle ABC$ has vertex B. For lines ℓ, m in the oriented Euclidean plane, $\angle(\ell, m)$ denotes their directed angle modulo π. In angle-chasing arguments, directed angles eliminate the need to distinguish many configurations involving interior and exterior angles.

Theorem 3.2. Triangle Inequality
The side lengths $a, b, c > 0$ of a nondegenerate triangle satisfy

$$a < b + c, \qquad b < c + a, \qquad c < a + b.$$

Conversely, any three positive numbers satisfying these inequalities are side lengths of a nondegenerate Euclidean triangle.

Theorem 3.3. Triangle Angle Sum and Exterior Angle Theorem
The interior angles of a nondegenerate Euclidean triangle satisfy

$$A + B + C = 180^\circ.$$

An exterior angle of a triangle equals the sum of its two remote interior angles.

Theorem 3.4. Congruence and Similarity Criteria
Two Euclidean triangles are congruent under SSS, SAS, or ASA (equivalently AAS). Two triangles are similar under AA, proportional corresponding sides (SSS similarity), or SAS similarity. For similar triangles with corresponding-side ratio $k > 0$, the ratios of their perimeters and areas are k and k^2, respectively.

Formula 3.5. Laws of Sines and Cosines
For $\triangle ABC$, write $a = BC$, $b = CA$, $c = AB$, and let R be its circumradius. Then

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R, \qquad a^2 = b^2 + c^2 - 2bc\cos A.$$

The analogous cosine-law identities follow by cyclic permutation.

Formula 3.6. Regular Polygons and Circles
For a regular n-gon ($n \geq 3$) with side length a, circumradius R, and apothem r,

$$a = 2R\sin\frac{\pi}{n}, \qquad r = R\cos\frac{\pi}{n}, \qquad \text{Area} = \frac{na^2}{4\tan(\pi/n)}.$$

A circle of radius R has circumference $2\pi R$ and area πR^2.

Formula 3.7. Triangle Area Formulas
If $\Delta = [ABC]$ and $s = (a+b+c)/2$, then

$$\Delta = \frac{1}{2}bc\sin A = rs = \frac{abc}{4R}, \qquad \Delta^2 = s(s-a)(s-b)(s-c),$$

where r is the inradius. If h_a is the altitude to the side of length a, then $\Delta = ah_a/2$.

Theorem 3.8. Angle Bisector Theorem
If the internal bisector of $\angle BAC$ meets BC at D, then

$$\frac{BD}{DC} = \frac{AB}{AC}.$$

Conversely, if D is interior to BC and satisfies this ratio, then AD bisects $\angle BAC$. The external bisector has the analogous ratio for external division, when defined.

Theorem 3.9. Stewart's Theorem
Let $D \in \overline{BC}$, with $BD = m$, $DC = n$, $BC = a = m+n$, and $AB = c$, $AC = b$, $AD = d$. Then

$$b^2m + c^2n = a(d^2 + mn).$$

Formula 3.10. Median and Angle-Bisector Lengths
If m_a is the median from A and ℓ_a is the internal angle bisector from A, then

$$m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}, \qquad \ell_a^2 = bc\left(1 - \frac{a^2}{(b+c)^2}\right).$$

Theorem 3.11. Ceva's Theorem
Let D, E, F lie in the interiors of BC, CA, AB, respectively. The cevians AD, BE, CF are concurrent if and only if

$$\frac{BD}{DC}\frac{CE}{EA}\frac{AF}{FB} = 1.$$

For points on the extended side lines, the corresponding directed-ratio formulation applies.

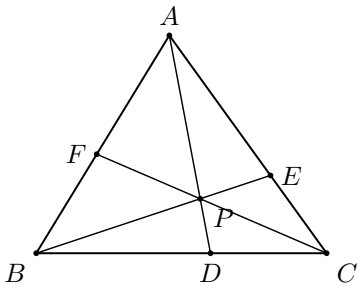


Figure 3.1: Three concurrent cevians in $\triangle ABC$.

Theorem 3.12. Menelaus' Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively, with none a vertex and no two of D, E, F coinciding. Suppose an odd number of D, E, F lie outside their corresponding closed sides. Then D, E, F are collinear if and only if

$$\frac{BD}{DC}\frac{CE}{EA}\frac{AF}{FB} = 1,$$

where the numerator and denominator are ordinary (unsigned) lengths. In a collinear configuration with no vertex intersections, the number of exterior points is necessarily odd.

Theorem 3.13. Trigonometric Ceva
For interior cevians AD, BE, CF of $\triangle ABC$,

$$\frac{\sin\angle BAD}{\sin\angle DAC}\frac{\sin\angle CBE}{\sin\angle EBA}\frac{\sin\angle ACF}{\sin\angle FCB} = 1$$

if and only if AD, BE, CF are concurrent.

Definition 3.14. Classical Triangle Centers
The *circumcenter* O is the intersection of the perpendicular bisectors; the *incenter* I is the intersection of the internal angle bisectors; the *centroid* G is the intersection of the medians; and the *orthocenter* H is the intersection of the altitudes. The circumcenter and orthocenter need not lie inside the triangle.

Theorem 3.15. Centroid Ratio
The three medians of a triangle meet at G. If D is the midpoint of BC, then $AG : GD = 2 : 1$. With position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ for the vertices,

$$\mathbf{g} = \frac{\mathbf{a}+\mathbf{b}+\mathbf{c}}{3}.$$

Theorem 3.16. Euler Line
For a triangle with circumcenter O, centroid G, and orthocenter H,

$$\overrightarrow{OH} = 3\overrightarrow{OG}.$$

Consequently, unless the three points coincide, they are collinear and $OG : GH = 1 : 2$.

Theorem 3.17. Orthocenter Reflection Lemma
Let H be the orthocenter of $\triangle ABC$. The reflection of H across each sideline lies on the circumcircle of $\triangle ABC$. The reflection of H across the midpoint of any side also lies on that circumcircle.

Theorem 3.18. Nine-Point Circle
The midpoints of the sides, the feet of the altitudes, and the midpoints of the segments joining the orthocenter to the vertices lie on one circle. Its center is the midpoint of OH, and its radius is $R/2$.

Formula 3.19. Euler's Incenter–Circumcenter Formula
If O, I are the circumcenter and incenter of a triangle, then

$$OI^2 = R(R - 2r).$$

In particular, $R \geq 2r$, with equality exactly for an equilateral triangle.

Definition 3.20. Excenters and Excircles
The three external angle bisectors, together with the appropriate internal bisector, determine the excenters I_a, I_b, I_c. The A-excircle has radius r_a and is tangent to BC and the extensions of AB, AC.

Formula 3.21. Exradii and Contact Lengths
For the exradii r_a, r_b, r_c of $\triangle ABC$,

$$\Delta = r_a(s-a) = r_b(s-b) = r_c(s-c).$$

If the incircle touches BC, CA, AB at D, E, F, respectively, then

$$BD = BF = s - b, \quad CD = CE = s - c, \quad AE = AF = s - a.$$

Definition 3.22. Isogonal Lines and Isogonal Conjugates
Two lines through A are *isogonal* in $\angle BAC$ if they are reflections of each other in its angle bisector. Two points P, Q not on the sidelines of $\triangle ABC$ are *isogonal conjugates* when AP, AQ are isogonal at A, and likewise at B and C.

Theorem 3.23. Symmedian Lemma
Let $K \in BC$ and suppose AK is the isogonal of the median from A. Then

$$\frac{BK}{KC} = \frac{AB^2}{AC^2}.$$

The three symmedians are concurrent at the symmedian point (Lemoine point).

Theorem 3.24. Gergonne and Nagel Points
The cevians from the vertices to the points of tangency of the incircle with the opposite sides are concurrent at the Gergonne point*. The cevians to the points of tangency of the corresponding excircles with the opposite sides are concurrent at the* Nagel point*.*

Theorem 3.25. Feuerbach's Theorem
The nine-point circle of a non-equilateral triangle is internally tangent to its incircle and externally tangent to each of its three excircles. For an equilateral triangle, the nine-point circle and incircle coincide.

Theorem 3.26. Fermat–Torricelli Point
If all angles of $\triangle ABC$ are less than 120°, there is a unique point P minimizing $PA+PB+PC$, characterized by

$$\angle APB = \angle BPC = \angle CPA = 120^\circ.$$

If one angle is at least 120°, its vertex minimizes that sum.

Formula 3.27. Half-Angle Relations in a Triangle
For $s = (a+b+c)/2$ and inradius r,

$$\tan\frac{A}{2} = \frac{r}{s-a}, \qquad \sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}},$$

with cyclic analogues. Moreover,

$$r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$$

Theorem 3.28. Morley's Trisector Theorem
The three points obtained by intersecting pairs of adjacent internal angle trisectors of any nondegenerate triangle form an equilateral triangle.

Theorem 3.29. Directed-Angle Criterion for Concyclicity
Let A, B, C be noncollinear points, and let D be distinct from them. The four points are concyclic if and only if

$$\angle(\overleftrightarrow{AB}, \overleftrightarrow{AC}) \equiv \angle(\overleftrightarrow{DB}, \overleftrightarrow{DC}) \pmod{\pi},$$

where the directed angles use the lines determined by the indicated pairs of points.

Theorem 3.30. Inscribed Angle and Tangent–Chord Theorems
An inscribed angle subtending an arc has half the measure of the corresponding central angle. The angle between a tangent to a circle at A and a chord AB equals the inscribed angle subtending chord AB, with directed angles interpreted modulo π.

Theorem 3.31. Cyclic Quadrilateral Criterion
A convex quadrilateral $ABCD$ is cyclic if and only if

$$\angle ABC + \angle ADC = 180^\circ.$$

Equivalently, its opposite angles are supplementary. With directed angles, cyclicity may be characterized without assuming convexity.

Theorem 3.32. Power of a Point
For a circle ω with center O and radius R, define

$$\mathrm{Pow}(P, \omega) = PO^2 - R^2.$$

If a line through P intersects the circle at A, B, let t_A, t_B be the signed coordinates of A, B along that line, measured from P in a common orientation. Then

$$t_A t_B = \mathrm{Pow}(P, \omega).$$

In particular, for external secants, the ordinary positive lengths satisfy $PA \cdot PB = PC \cdot PD$. For a tangent PT,

$$PT^2 = \mathrm{Pow}(P, \omega).$$

Theorem 3.33. Radical Axis and Radical Center
For two distinct nonconcentric circles ω_1, ω_2, the locus of points satisfying

$$\mathrm{Pow}(P, \omega_1) = \mathrm{Pow}(P, \omega_2)$$

is a line perpendicular to their line of centers, called the radical axis. *For three circles whose centers are not collinear, the pairwise radical axes are concurrent at their* radical center.

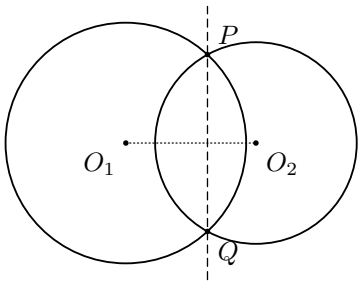


Figure 3.2: The radical axis of two intersecting circles is their common chord line.

Theorem 3.34. Ptolemy's Theorem and Inequality
For a convex cyclic quadrilateral $ABCD$,

$$AC \cdot BD = AB \cdot CD + BC \cdot DA.$$

For any four points in the Euclidean plane, the corresponding inequality

$$AC \cdot BD \leq AB \cdot CD + BC \cdot DA$$

holds. For four distinct vertices in convex order, equality characterizes cyclicity.

Formula 3.35. Brahmagupta's Formula
For a convex cyclic quadrilateral with side lengths a, b, c, d and semiperimeter $s = (a+b+c+d)/2$, its area K satisfies

$$K = \sqrt{(s-a)(s-b)(s-c)(s-d)}.$$

Theorem 3.36. Pitot's Theorem
If a convex quadrilateral $ABCD$ has an incircle tangent to all four sides, then

$$AB + CD = BC + DA.$$

This equality alone does not characterize tangential quadrilaterals without additional hypotheses.

Theorem 3.37. Varignon's Theorem
The midpoints of the sides of any quadrilateral form a parallelogram. Its area is half the area of the original quadrilateral when the original quadrilateral is simple.

Theorem 3.38. Simson Line
A point P lies on the circumcircle of a triangle ABC if and only if its perpendicular projections onto the three side lines AB, BC, CA are collinear, provided the triangle is nondegenerate. Their common line is called the Simson line *of P.*

Theorem 3.39. Reim's Theorem
Let two circles meet at distinct points P, Q. Suppose a line through P meets the circles again at A, B, and a line through Q meets them again at C, D, with A, C on the first circle and B, D on the second. Then $AC \parallel BD$.

Theorem 3.40. Miquel's Theorem
Let $D \in BC$, $E \in CA$, and $F \in AB$ be points distinct from the vertices of $\triangle ABC$. The three circumcircles of AEF, BFD, and CDE pass through a common point (their Miquel point*), apart from possible pairwise shared vertices.*

Definition 3.41. Direct Similarity and Spiral Similarity
An orientation-preserving similarity of the complex plane has the form

$$z \longmapsto az + b, \qquad a \in \mathbb{C} \setminus \{0\},\ b \in \mathbb{C}.$$

If $a \neq 1$, it has a unique fixed point $z_0 = b/(1-a)$ and is a rotation followed by a homothety centered at z_0. If $a = 1$, it is a translation. The former case is commonly called a *spiral similarity*.

Definition 3.42. Homothety
For a point O and a real scalar $k \neq 0$, the homothety with center O and ratio k maps a point X to X' satisfying

$$\overrightarrow{OX'} = k\overrightarrow{OX}.$$

It multiplies all distances by $|k|$ and areas by k^2.

Definition 3.43. Inversion in a Circle
Fix a circle with center O and radius $r > 0$. The inversion sends each $P \neq O$ to P' on the ray OP such that

$$OP \cdot OP' = r^2.$$

It is an involution. A generalized circle (a circle or a line) not passing through O is mapped to a generalized circle, and a circle through O is mapped to a line not through O.

Proposition 3.44. Inversion Distance Identity
Under inversion with center O and radius r, for $P, Q \neq O$ with images P', Q',

$$P'Q' = \frac{r^2}{OP \cdot OQ} PQ.$$

Inversion preserves the magnitude of angles between intersecting smooth curves, but reverses their orientation.

Theorem 3.45. Butterfly Theorem
Let M be the midpoint of a chord PQ of a circle. Let chords AB and CD pass through M, with A, C on one side of PQ and B, D on the other. If AD and BC meet the line PQ at X and Y, respectively, then M is the midpoint of XY.

Proposition 3.46. Circles Orthogonal to an Inversion Circle
A circle ω not passing through the center O of an inversion circle is invariant as a set under that inversion if and only if it is orthogonal to the inversion circle, meaning that their tangent lines at either intersection point are perpendicular.

Theorem 3.47. Desargues' Theorem
In the real projective plane, if two triangles have corresponding vertices joined by three concurrent lines, then the intersections of corresponding sides are collinear. Conversely, if the three side intersections are collinear, then the lines joining corresponding vertices are concurrent in the projective plane.

Theorem 3.48. Pappus' Theorem
Let A, B, C lie on one line and A', B', C' on another, with all pairwise intersections below defined. Then the three points

$$AB' \cap A'B, \quad AC' \cap A'C, \quad BC' \cap B'C$$

are collinear in the real projective plane.

Theorem 3.49. Pascal's Theorem
For six distinct points A, B, C, D, E, F on a nondegenerate conic, the three intersections of opposite sides

$$AB \cap DE, \qquad BC \cap EF, \qquad CD \cap FA$$

are collinear in the projective plane, allowing points at infinity for parallel sides.

Definition 3.50. Cross Ratio
For four distinct collinear points A, B, C, D on a projective line, their cross ratio in an affine coordinate t is

$$(A, B; C, D) = \frac{(t_C - t_A)(t_D - t_B)}{(t_C - t_B)(t_D - t_A)},$$

with the usual limiting interpretation when one point is at infinity. Projective transformations preserve the cross ratio.

Formula 3.51. Coordinate and Shoelace Area Formulas
For $A = (x_1, y_1)$, $B = (x_2, y_2)$, $C = (x_3, y_3)$,

$$[ABC] = \frac{1}{2} \left| \det \begin{pmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{pmatrix} \right|.$$

For a simple polygon with cyclically ordered vertices (x_i, y_i), $1 \le i \le n$,

$$\text{Area} = \frac{1}{2}\left|\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1}y_i)\right|, \quad (x_{n+1}, y_{n+1}) = (x_1, y_1).$$

Definition 3.52. Barycentric Coordinates
For a nondegenerate triangle with position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$, a point P has barycentric coordinates $(\alpha : \beta : \gamma)$ when

$$\mathbf{p} = \frac{\alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}}{\alpha + \beta + \gamma}, \qquad \alpha + \beta + \gamma \neq 0.$$

The coordinates are homogeneous: multiplying all three by a common nonzero scalar does not change the point.

Proposition 3.53. Barycentric Coordinates of Triangle Centers
Relative to $\triangle ABC$, the centroid and incenter have barycentric coordinates

$$G = (1 : 1 : 1), \qquad I = (a : b : c),$$

where $a = BC$, $b = CA$, $c = AB$. The symmedian point has coordinates $(a^2 : b^2 : c^2)$.

Formula 3.54. Complex Coordinates on the Unit Circle
For distinct points $a, b, c, d \in \mathbb{C}$ with $|a| = |b| = |c| = |d| = 1$,

$$\frac{a-b}{c-d} = \frac{ab}{cd}\overline{\left(\frac{a-b}{c-d}\right)}.$$

If a, b, c, d are four distinct points on the unit circle, then their complex cross ratio

$$\frac{(a-c)(b-d)}{(a-d)(b-c)}$$

is real.

Theorem 3.55. Pick's Theorem
For a simple lattice polygon in $\mathbb{R}^2$, with I lattice points in its interior and B lattice points on its boundary,

$$\text{Area} = I + \frac{B}{2} - 1.$$

Theorem 3.56. Helly's Theorem in the Plane
Let $\mathcal{F}$ be a finite family of convex subsets of $\mathbb{R}^2$ with at least three members. If every three members have a common point, then all members have a common point.

Theorem 3.57. Carathéodory's Theorem in the Plane
If x belongs to the convex hull of a set $S \subseteq \mathbb{R}^2$, then x is a convex combination of at most three points of S.

Theorem 3.58. Isoperimetric Inequality in the Plane
Let a simple closed rectifiable curve in the plane have length L and enclose area A. Then

$$L^2 \ge 4\pi A,$$

with equality if and only if the curve is a circle.

Formula 3.59. Basic Solid Geometry
A right circular cylinder of radius r and height h has volume $\pi r^2 h$; a right circular cone has volume $\pi r^2 h/3$; and a ball of radius r has volume $4\pi r^3/3$ and boundary area $4\pi r^2$.

Theorem 3.60. Euler's Polyhedron Formula
For a convex polyhedron with V vertices, E edges, and F faces,

$$V - E + F = 2.$$

The same identity holds for a connected planar graph in a plane embedding when the unbounded face is included.

Theorem 3.61. Cavalieri's Principle
Two measurable plane regions having equal cross-sectional lengths at almost every level in a fixed direction have equal area, provided their areas are finite. Similarly, solids with equal cross-sectional areas at almost every height have equal volume when the volumes are finite.

Chapter 4

Combinatorics and Probability

Theorem 4.1. Addition and Multiplication Principles
If mutually exclusive classes of objects have respective cardinalities $a_1, \ldots, a_k$, their union has cardinality $a_1 + \cdots + a_k$. If an object is determined by k independent choices with a_i possibilities at step i, the number of objects is $a_1 \cdots a_k$.

Definition 4.2. Factorials, Permutations, and Combinations
For $n \in \mathbb{N}_0$, let $n! = \prod_{k=1}^{n} k$, with $0! = 1$. If $0 \leq r \leq n$, the number of ordered selections of r distinct elements from an n-element set is

$$_nP_r = \frac{n!}{(n-r)!},$$

and the number of r-element subsets is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Formula 4.3. Multiset Combinations and Stars and Bars
For $n \geq 1$ and $r \geq 0$, the number of multisets of cardinality r chosen from n types is

$$\left(\!\binom{n}{r}\!\right) = \binom{n+r-1}{r}.$$

Equivalently, this is the number of nonnegative integer solutions of $x_1 + \cdots + x_n = r$. The number of positive integer solutions for $r \geq n$ is $\binom{r-1}{n-1}$.

Theorem 4.4. Binomial and Multinomial Theorems
For $n \in \mathbb{N}_0$,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

If $n_1, \ldots, n_m \in \mathbb{N}_0$ sum to n, the corresponding multinomial coefficient is

$$\binom{n}{n_1, \ldots, n_m} = \frac{n!}{n_1! \cdots n_m!},$$

and $(x_1 + \cdots + x_m)^n$ expands as the sum of these coefficients times $x_1^{n_1} \cdots x_m^{n_m}$.

Formula 4.5. Basic Binomial Identities
For $0 \le k \le n$,

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k},$$

where coefficients with inadmissible lower indices are zero. Furthermore,

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n, \quad \sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0 \quad (n \ge 1),$$

$$\sum_{k=0}^{n} k \binom{n}{k} = n 2^{n-1} \quad (n \ge 1).$$

Theorem 4.6. Vandermonde's Identity
For nonnegative integers r, s, n,

$$\sum_{k=0}^{n} \binom{r}{k} \binom{s}{n-k} = \binom{r+s}{n}.$$

In particular,

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}.$$

Theorem 4.7. Binomial Inversion
For sequences (a_n) and (b_n) over a field of characteristic zero,

$$b_n = \sum_{k=0}^{n} \binom{n}{k} a_k \quad \Longleftrightarrow \quad a_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} b_k.$$

Theorem 4.8. Pigeonhole Principle
If N objects are placed in $k \ge 1$ boxes, some box contains at least $\lceil N/k \rceil$ objects. In particular, placing $k+1$ objects in k boxes forces at least two into the same box.

Theorem 4.9. Inclusion–Exclusion Principle
For finite sets $A_1, \ldots, A_n$,

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{\varnothing \ne S \subseteq [n]} (-1)^{|S|+1} \left| \bigcap_{i \in S} A_i \right|.$$

Formula 4.10. Counting Surjections
For $n \ge 0$ and $k \ge 1$, the number of surjections from an n-element set to a k-element set is

$$\sum_{j=0}^{k} (-1)^j \binom{k}{j} (k-j)^n = k!\, S(n,k),$$

where $S(n,k)$ is the Stirling number of the second kind and $0^0 = 1$ in the finite counting sum.

Definition 4.11. Derangement
A derangement is a permutation with no fixed point. Write D_n or $!n$ for the number of derangements of n elements; here D_n is not the dihedral group, the notation being determined by context.

Formula 4.12. Derangement Formula
For $n \in \mathbb{N}_0$,

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

For $n \geq 2$, $D_n = (n-1)(D_{n-1} + D_{n-2})$, with $D_0 = 1$ and $D_1 = 0$. For $n \geq 1$, D_n is the nearest integer to $n!/e$.

Definition 4.13. Stirling Numbers and Bell Numbers
The unsigned Stirling number $c(n, k)$ counts permutations of $[n]$ with k cycles. The Stirling number $S(n, k)$ counts partitions of $[n]$ into k nonempty blocks. The Bell number

$$B_n = \sum_{k=0}^{n} S(n, k)$$

counts all set partitions of $[n]$.

Formula 4.14. Stirling Number Recurrences
For $n, k \geq 1$,

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k),$$
$$S(n, k) = S(n-1, k-1) + kS(n-1, k).$$

The boundary conditions are $c(0, 0) = S(0, 0) = 1$ and zero for inadmissible indices.

Definition 4.15. Catalan Numbers
For $n \in \mathbb{N}_0$, the nth Catalan number is

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1}.$$

It counts, among other objects, correctly matched parenthesis strings of length $2n$ and lattice paths from $(0, 0)$ to (n, n) that never rise above the diagonal.

Formula 4.16. Catalan Recurrence and Generating Function
The Catalan numbers satisfy

$$C_0 = 1, \qquad C_{n+1} = \sum_{k=0}^{n} C_k C_{n-k}.$$

Their ordinary generating function is

$$C(x) = \sum_{n \geq 0} C_n x^n = \frac{1 - \sqrt{1 - 4x}}{2x},$$

with the constant term understood by formal-power-series expansion.

Definition 4.17. Integer Partitions and Compositions
A partition of $n \in \mathbb{N}_0$ is a weakly decreasing sequence of positive integers summing to n, with the empty partition for $n = 0$. A composition of $n \in \mathbb{N}$ is an ordered sequence of positive integers summing to n. The number of compositions of n is 2^{n-1}.

Theorem 4.18. Euler's Partition Generating Function
If $p(n)$ is the number of integer partitions of n with $p(0) = 1$, then, as a formal power series,

$$\sum_{n\geq 0} p(n)x^n = \prod_{k\geq 1} \frac{1}{1-x^k}.$$

The number of partitions into distinct parts equals the number of partitions into odd parts.

Theorem 4.19. Reflection Principle for Lattice Paths
For nonnegative integers a, b with $a \geq b$, the number of lattice paths from $(0,0)$ to (a,b) using right and up steps and never going above $y = x$ is

$$\binom{a+b}{b} - \binom{a+b}{b-1} = \frac{a-b+1}{a+1}\binom{a+b}{b}.$$

For $a = b = n$, this equals C_n.

Definition 4.20. Recurrence Relation and Generating Function
A recurrence relation specifies sequence terms in terms of earlier terms. The ordinary generating function of $(a_n)_{n\geq 0}$ is

$$A(x) = \sum_{n\geq 0} a_n x^n,$$

viewed either formally or as an analytic series where it converges. The exponential generating function is $\sum_{n\geq 0} a_n x^n/n!$.

Formula 4.21. Geometric and Binomial Series
As formal power series, for $m \in \mathbb{N}$,

$$\frac{1}{1-x} = \sum_{n\geq 0} x^n, \qquad \frac{1}{(1-x)^m} = \sum_{n\geq 0} \binom{n+m-1}{m-1} x^n.$$

The same identities hold analytically when $|x| < 1$.

Theorem 4.22. Generating Functions for Linear Recurrences
A sequence satisfying a linear recurrence with constant coefficients for all sufficiently large indices has a rational ordinary generating function. Conversely, the coefficients of any rational formal power series satisfy a constant-coefficient linear recurrence for all sufficiently large indices.

Formula 4.23. Coefficient Extraction and Convolution
For formal power series $A(x) = \sum_{n\geq 0} a_n x^n$ and $B(x) = \sum_{n\geq 0} b_n x^n$,

$$[x^n](A(x)B(x)) = \sum_{k=0}^{n} a_k b_{n-k}.$$

More generally, $[x^n]x^k A(x) = a_{n-k}$ for $n \geq k$.

Theorem 4.24. Lagrange Inversion, Basic Form
Let $\phi(t)$ be a formal power series over a field of characteristic zero with $\phi(0) \neq 0$. If $T(x)$ is the unique formal series with $T(0) = 0$ satisfying

$$T = x\phi(T),$$

then for $n \geq 1$,

$$[x^n]T(x) = \frac{1}{n}[t^{n-1}]\phi(t)^n.$$

Theorem 4.25. Burnside's Lemma
If a finite group G acts on a finite set X, the number of orbits is

$$|G\backslash X| = \frac{1}{|G|}\sum_{g\in G} |\operatorname{Fix}(g)|,$$

where $\operatorname{Fix}(g) = \{x \in X \mid g \cdot x = x\}$.

Theorem 4.26. Pólya Enumeration Theorem
Let a finite group G permute an n-element set, and suppose each position receives one of $q \in \mathbb{N}$ colors. If $c(g)$ denotes the number of cycles of g in its permutation of the positions, the number of colorings up to G-symmetry is

$$\frac{1}{|G|}\sum_{g\in G} q^{c(g)}.$$

Definition 4.27. Graphs, Degrees, and Paths
A (finite simple undirected) graph is $G = (V, E)$, where E consists of unordered pairs of distinct vertices. The degree $\deg(v)$ is the number of edges incident with v. A path has no repeated vertices, while a cycle is a closed path with at least three vertices and no repeated internal vertices.

Theorem 4.28. Handshaking Lemma
For any finite undirected graph,

$$\sum_{v\in V(G)} \deg(v) = 2|E(G)|.$$

Consequently, the number of vertices of odd degree is even.

Definition 4.29. Trees and Spanning Trees
A tree is a connected graph with no cycles. A spanning tree of a connected graph is a subgraph that is a tree and contains every vertex.

Theorem 4.30. Equivalent Characterizations of a Tree
For a finite simple graph G with $n \geq 1$ vertices, any two of the following properties imply the third: G is connected; G is acyclic; G has exactly $n-1$ edges. In particular, a tree on n vertices has $n-1$ edges and there is a unique simple path between any two of its vertices.

Theorem 4.31. Cayley's Formula
For $n \geq 2$, the number of labeled trees on vertex set $[n]$ is

$$n^{n-2}.$$

For $n = 1$, there is exactly one such tree.

Definition 4.32. Bipartite Graph
A graph $G = (V, E)$ is bipartite if V can be partitioned into disjoint sets U, W so that every edge joins a vertex of U to a vertex of W.

Theorem 4.33. Bipartite Graph Criterion
A finite graph is bipartite if and only if it has no cycle of odd length.

Definition 4.34. Vertex Coloring and Chromatic Number
A proper vertex coloring of a graph G assigns colors to the vertices so that adjacent vertices receive distinct colors. The *chromatic number* $\chi(G)$ is the smallest number of colors in a proper coloring.

Theorem 4.35. Greedy Coloring Bound
Every finite simple graph G satisfies

$$\chi(G) \le \Delta(G) + 1,$$

where $\Delta(G)$ is its maximum vertex degree. Every bipartite graph with at least one edge has chromatic number 2.

Theorem 4.36. Dirac's and Ore's Hamiltonian Criteria
Let G be a finite simple graph with $n \ge 3$ vertices. If every vertex has degree at least $n/2$, then G has a Hamiltonian cycle (Dirac). More generally, if $\deg(u) + \deg(v) \ge n$ for every pair of distinct nonadjacent vertices u, v, then G has a Hamiltonian cycle (Ore).

Theorem 4.37. Eulerian Trail and Circuit Criteria
A finite connected graph has an Eulerian circuit exactly when every vertex has even degree. It has an Eulerian trail with two distinct endpoints exactly when precisely two vertices have odd degree.

Theorem 4.38. Euler's Formula for Planar Graphs
If a finite connected graph has a plane embedding with V vertices, E edges, and F faces, including the unbounded face, then

$$V - E + F = 2.$$

For a simple planar graph with $V \ge 3$,

$$E \le 3V - 6.$$

If it is also bipartite, then $E \le 2V - 4$.

Theorem 4.39. Kuratowski's Theorem
A finite graph is planar if and only if it contains no subdivision of K_5 or $K_{3,3}$ as a subgraph.

Definition 4.40. Matching and Vertex Cover
A matching is a set of edges no two of which share a vertex. A matching is perfect if every vertex is incident with a matching edge. A vertex cover is a set of vertices incident with every edge.

Theorem 4.41. Hall's Marriage Theorem
A finite bipartite graph with vertex classes U, W has a matching saturating every vertex of U if and only if

$$|N(S)| \ge |S| \qquad \text{for every } S \subseteq U,$$

where $N(S)$ is the set of neighbors of vertices in S.

Theorem 4.42. Kőnig's Theorem
In every finite bipartite graph, the maximum size of a matching equals the minimum size of a vertex cover.

Theorem 4.43. Tutte's Perfect Matching Theorem
A finite undirected graph G has a perfect matching if and only if, for every $S \subseteq V(G)$, the graph $G - S$ has at most $|S|$ connected components of odd order.

Theorem 4.44. Max-Flow Min-Cut Theorem
In a finite directed capacitated network with nonnegative edge capacities and distinguished source s and sink t, the maximum possible value of an s–t flow equals the minimum capacity of an s–t cut.

Theorem 4.45. Menger's Theorem, Vertex Form
For distinct nonadjacent vertices u, v in a finite graph, the maximum number of internally vertex-disjoint u–v paths equals the minimum number of vertices whose removal separates u and v.

Theorem 4.46. Matrix–Tree Theorem
Let G be a finite connected graph with adjacency matrix $\mathbf{A}$ and degree matrix $\mathbf{D}$. Set $\mathbf{L} = \mathbf{D} - \mathbf{A}$. The number of spanning trees of G equals any principal cofactor of $\mathbf{L}$.

Definition 4.47. Partially Ordered Sets, Chains, and Antichains
A partially ordered set (poset) is a set X with a reflexive, antisymmetric, transitive relation $\preceq$. A chain is a subset whose elements are pairwise comparable; an antichain is a subset whose distinct elements are pairwise incomparable.

Theorem 4.48. Dilworth's Theorem
In a finite poset, the maximum cardinality of an antichain equals the minimum number of chains in a partition of the poset.

Theorem 4.49. Sperner's Theorem
For a family $\mathcal{A}$ of subsets of $[n]$, no one of which properly contains another,

$$|\mathcal{A}| \leq \binom{n}{\lfloor n/2 \rfloor}.$$

Theorem 4.50. Lubell–Yamamoto–Meshalkin Inequality
If $\mathcal{A}$ is an antichain in the Boolean lattice $2^{[n]}$, then

$$\sum_{A \in \mathcal{A}} \frac{1}{\binom{n}{|A|}} \leq 1.$$

Theorem 4.51. Ramsey's Theorem
For positive integers $r, k_1, \ldots, k_m$, there exists N such that every m-coloring of the r-element subsets of $[N]$ contains a set of size k_i all of whose r-element subsets have color i, for at least one index i. In the graph case, $R(s,t)$ is the least N such that every red–blue edge coloring of K_N has a red K_s or a blue K_t.

Theorem 4.52. Erdős–Szekeres Monotone Subsequence Theorem
Every sequence of $(r-1)(s-1)+1$ distinct real numbers contains an increasing subsequence of length r or a decreasing subsequence of length s.

Theorem 4.53. Turán's Theorem
Among simple graphs on n vertices containing no complete graph K_{r+1}, the unique graph with the maximum number of edges (up to isomorphism) is the complete r-partite graph whose part sizes differ by at most one.

Definition 4.54. Probability Space and Conditional Probability
A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where $\mathcal{F}$ is a σ-algebra of subsets of Ω and $\mathbb{P}$ is a countably additive probability measure with $\mathbb{P}(\Omega) = 1$. For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For a finite uniform space Ω, $\mathbb{P}(A) = |A|/|\Omega|$.

Theorem 4.55. Addition Rule and Union Bound
For events A, B,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

For any finite or countable family of events,

$$\mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i).$$

Theorem 4.56. Law of Total Probability and Bayes' Theorem
Let $(B_i)_{i=1}^m$ be a partition of Ω into events of positive probability. For any event A,

$$\mathbb{P}(A) = \sum_{i=1}^{m} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

If $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{m} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

Definition 4.57. Independence
Events A, B are independent when $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. A family of events is mutually independent if this product identity holds for every finite nonempty subfamily. Random variables X, Y are independent if the events $\{X \in U\}$ and $\{Y \in V\}$ are independent for all Borel sets U, V.

Definition 4.58. Random Variable and Expectation
A real-valued random variable X is a measurable map from Ω to $\mathbb{R}$. Its expectation, when defined, is denoted $\mathbb{E}[X]$. For a discrete random variable with countable range and $\sum_x |x|\mathbb{P}(X = x) < \infty$,

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x).$$

Definition 4.59. Distribution of a Random Variable
The distribution of X is specified by $\mathbb{P}(X \in B)$ for Borel sets B. For a discrete random variable, its probability mass function is $p_X(x) = \mathbb{P}(X = x)$; its cumulative distribution function is $F_X(x) = \mathbb{P}(X \leq x)$.

Formula 4.60. Bernoulli, Binomial, and Geometric Distributions
If $X \sim \mathrm{Bern}(p)$, then $\mathbb{P}(X = 1) = p$ and $\mathbb{E}[X] = p$, $\mathrm{Var}(X) = p(1-p)$. If $X \sim \mathrm{Bin}(n, p)$, then for $0 \le k \le n$,

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \mathbb{E}[X] = np, \quad \mathrm{Var}(X) = np(1-p).$$

If $Y \sim \mathrm{Geom}(p)$ counts the number of independent Bernoulli trials up to and including the first success, with $0 < p \le 1$, then

$$\mathbb{P}(Y = k) = (1-p)^{k-1} p \quad (k \in \mathbb{N}), \qquad \mathbb{E}[Y] = \frac{1}{p}.$$

Formula 4.61. Hypergeometric Distribution
If a sample of size n is drawn uniformly without replacement from N objects, of which K are designated successes, the number X of successes satisfies

$$\mathbb{P}(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

for integers k with $\max(0, n - (N-K)) \le k \le \min(n, K)$, and $\mathbb{E}[X] = nK/N$.

Definition 4.62. Conditional Expectation
For integrable X and a random variable Y, the conditional expectation $\mathbb{E}[X \mid Y]$ denotes an integrable function of Y whose expectation on each event determined by Y agrees with that of X. In the discrete case, when $\mathbb{P}(Y = y) > 0$,

$$\mathbb{E}[X \mid Y = y] = \sum_x x \mathbb{P}(X = x \mid Y = y),$$

provided the sum is absolutely convergent.

Theorem 4.63. Linearity of Expectation
For integrable random variables X, Y and scalars a, b,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

No independence assumption is needed. If $X = \sum_{i=1}^{n} \mathbb{1}_{A_i}$ counts how many of the events A_i occur, then

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

Formula 4.64. Variance and Covariance
For square-integrable real random variables X, Y,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2,$$

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y).$$

If X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$.

Theorem 4.65. Law of Total Expectation and Total Variance
If X is integrable and Y is a random variable, then

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]].$$

If X is square-integrable, then

$$\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y]).$$

Theorem 4.66. Tail-Sum Formula
If X is a nonnegative integer-valued random variable, then, allowing the value $+\infty$,

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

Theorem 4.67. Markov's Inequality
If $X \geq 0$ almost surely and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

provided the right-hand side is finite.

Theorem 4.68. Chebyshev's Inequality
If X has finite variance and $t > 0$, then

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\operatorname{Var}(X)}{t^2}.$$

Theorem 4.69. Jensen's Inequality for Expectations
For an integrable random variable X taking values in an interval I and a convex function $f\colon I \to \mathbb{R}$, whenever the relevant expectations exist,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

Theorem 4.70. Chernoff Bounds for Binomial Random Variables
If $X \sim \operatorname{Bin}(n, p)$ has mean $\mu = np > 0$, then for $\delta > 0$,

$$\mathbb{P}(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu},$$

and for $0 < \delta < 1$,

$$\mathbb{P}(X \leq (1-\delta)\mu) \leq e^{-\mu\delta^2/2}.$$

Theorem 4.71. The First-Moment Method
If X is a nonnegative integer-valued random variable and $\mathbb{E}[X] < 1$, then $\mathbb{P}(X = 0) > 0$. More generally, if $\mathbb{E}[X] > 0$ and $X \geq 0$, then $\mathbb{P}(X > 0) > 0$.

Formula 4.72. Uniform Random Permutations
If π is uniformly distributed over the permutations of $[n]$, then the expected number of fixed points is 1 for $n \geq 1$, and the expected number of cycles is

$$H_n = \sum_{k=1}^{n} \frac{1}{k}.$$

The probability that π is a derangement is $D_n/n!$.

Formula 4.73. Coupon Collector
If one independently samples from n equally likely types until all types have appeared, the expected number of samples is

$$nH_n = n\sum_{k=1}^{n}\frac{1}{k}.$$

Formula 4.74. Gambler's Ruin Probability
For a nearest-neighbor random walk on $\{0, 1, \ldots, N\}$ absorbed at $0, N$, with independent steps $+1$ of probability p and -1 of probability $q = 1-p$, the probability of reaching N before 0 from $i \in \{0, \ldots, N\}$ is

$$\mathbb{P}_i(\text{hit } N \text{ before } 0) = \begin{cases} \dfrac{1-(q/p)^i}{1-(q/p)^N}, & p \neq q,\ 0 < p < 1, \\ \dfrac{i}{N}, & p = q = \frac{1}{2}. \end{cases}$$

At $p = 0$ or $p = 1$, the probability is determined by the deterministic walk.

Theorem 4.75. Lovász Local Lemma, Symmetric Form
Let $A_1, \ldots, A_n$ be events, each independent of all but at most d of the other events. If $\mathbb{P}(A_i) \leq p$ for all i and

$$ep(d+1) \leq 1,$$

then $\mathbb{P}(\bigcap_{i=1}^{n} A_i^c) > 0$.

Chapter 5

Calculus and Analysis

Chapter 6

Linear Algebra

Part II

Problem-Solving Strategies